



HUMANE Paper Review

How malicious AI swarms can threaten democracy

Daniel Thilo Schroeder¹, Meeyoung Cha², Andrea Baronchelli³, Nick Bostrom⁴, Nicholas A. Christakis⁵, David Garcia⁶, Amit Goldenberg⁷, Yara Kyrychenko⁸, Kevin Leyton-Brown⁹, Nina Lutz¹⁰, Gary Marcus¹¹, Filippo Menczer¹², Gordon Pennycook¹³, David G. Rand¹⁴, Maria Ressa^{15,16}, Frank Schweitzer¹⁷, Dawn Song¹⁸, Christopher Summerfield¹⁹, Audrey Tang²⁰, Jay J. Van Bavel^{11,21}, Sander van der Linden⁸, Jonas R. Kunst²²

송실대학교 문화콘텐츠학과, 석사과정생 이다현

Science

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Why this matters now

Advances in artificial intelligence (AI) offers the prospect of manipulating beliefs and behaviors on a population-wide level.

- LLMs + autonomous agents → unprecedented scale & precision
- Generative tools: more output without losing credibility
 - +) AI reasoning techniques (e.g., CoT prompting)
- Malicious AI Swarm: a **coordinated network of AI agents** that imitates human social behavior at scale
- Risk: persistent, adaptive manipulation

Background: democracy already vulnerable

- Erosion of rational-critical discourse
- Lack of shared reality
- Declining trust in core institutions (media, science, government)
- AI swarms as a potent accelerant
 - Impact not predetermined

What old ops looked like

- Human-driven botnets had limits: cost, cadence, iteration speed
- Example: IRA 2016 Twitter op
- Highly concentrated exposure
- Mixed interpretations of “effect” (e.g., sowing distrust)

드루킹이 조작한 주요 댓글

“대구 무시하면 더민주 지지도 철회한다. 추미애 파이팅”(stoc****)
“표리부동 이미지 코스프레 대명사...이재명시장인 듯”(kym0****)
“최재성 싸수가 노랑네~ 셀프좌표가 뭐니까?”(budd****)
“티브이에 나와 이미지 쇄신하는 연예인 말고 경기지사는 전해철의원이
쵸”(bell****) “안지사님 통합과 협치를 주장하는 소신으로 출마하면
잘 이끌어 갈 겁니다”(anma****)
“이재명+이명박=이재명박 아니냐~”(sung****)

What is a malicious AI swarm?

- Definition (5 properties)
 1. **Persistent** identities + memory
 2. Coordinated objectives, **varied tone/content**
 3. **Real-time adaptation** to signals & humans
 4. Minimal human oversight
 5. Cross-platform deployment

Why swarms are a step-change vs botnets

- Botnets: repetition, manual scheduling, rigid scripts
- Swarms: **scale + heterogeneity + adaptation**
- Capable of organic looking, context-aware content
- Can sustain coherent narratives across many agents

Capability: coordination, infiltration

1) From central command to fluid, real-time coordination

- Local adaptation + periodic synchronization
- Potential “hive” behavior / emergent group dynamics

2) Map networks & infiltrate communities

- Identify key communities/beliefs, track trends
- Tailor to cultural cues; win followers

Capability: evasion, optimization, persistence

3) Human-level mimicry to evade detectors

- Photorealistic avatars, slang, varied rhythms

4) Self-optimizing persuasion

- Micro A/B tests at machine speed

5) 24/7 presence → long-term infrastructure

- Gradual discourse shift; deeper cultural change

Harm pathway: synthetic consensus

- Fragmented environment + echo chambers
- Swarms can engineer synthetic consensus
- Create illusion of broad support (e.g., coordinated likes)
- People update beliefs via peer norms more than evidence
- Outcome: invisible, embedded nudging over time

Harm pathway: segmented realities & LLM grooming

- Segmented realities
 - Tailor misleading info to linguistic/cultural/emotional markers
 - Reduce cross-group consensus
- LLM grooming (data poisoning at scale)
 - Flood the web with fabricated chatter
 - Contaminate future training data → narratives “calcify” in model weights

Harm pathway: harassment + FUD + elite concentration

- Synthetic harassment drives targets out of public life
- FUD: “Much online speech may be AI-generated” → **disengagement**
- Content saturation → “epistemic vertigo”
- Overcompensation: retreat to trust proxies (followers, badges, fame)
→ elite/celebrity voices dominate; many-to-many shrinks

Defenses: raise attacker costs

*Design goal: **arms race** → increase attacker complexity, resources, discovery likelihood*

- First line: always-on detection + public audits
 - Focus on provenance/coordination, not semantic truth
- End-user: optional “AI shields”
 - Labels, down-rank/hide, short provenance explanations
 - Beware adversarial reporting
- Test: agent-based simulation to stress-test detectors

Defenses: raise attacker costs

*Design goal: **arms race** → increase attacker complexity, resources, discovery likelihood*

- Complement: strengthen provenance
 - Passkeys, cryptographic attestations, federated reputation
 - “Proof-of-human” not a panacea; protect anonymity
- Coordination: distributed “AI Influence Observatory”
- Economics: disrupt the commercial market for manipulation
 - No-revenue policies, discount synthetic engagement, audited bot-traffic metrics

Conclusion: the next few years are a window

- A proactive window before major political events
- If: detectors deployed + standardized persuasion stress-tests + open incident telemetry
- Requires coordination across science, civil society, industry, election security

Aim: mitigate risks without freezing innovation

Review

- 생성 콘텐츠가 고도로 발전해서, 내용의 진위를 따지는 것 보다 조작된 단체적 행동, 네트워크 패턴 등을 탐지하는게 중요하다고 함.
- 연구하고 있는 Stance detection이 중요할 수 있겠다고 생각함. 개별 텍스트의 편향을 자동 탐지하면 전체 여론의 흐름을 모니터링하는데 활용될 수 있기 때문.
- 텍스트를 넘어 이미지, 비디오 등 multimodal 데이터에 대한 Stance Detection이 더 중요해질 것 같음. 텍스트보다 사람들을 속이기 쉬운 modality 이며, 탐지 기술 자체도 덜 연구되었기 때문.

RQ

- 논문은 콘텐츠의 '진위 여부'를 판정하는게 아니라, 'coordination 여부'를 탐지해야 한다고 주장함. 어떻게 할 수 있을까?
- 논문은 웹이 오염되면 학습 데이터도 오염될 수 있다고 말함. 웹 오염이 모델의 어떤 능력을 가장 많이 무너뜨릴까? (factuality, calibration, bias 등)